Student Performance Prediction System

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1. Problem Description

The Loan Eligibility Prediction System is a machine learning-based application developed using Python to predict whether a person is eligible for a loan or likely to default.

Banks and financial institutions traditionally rely on manual verification of loan applications, which is time-consuming and prone to human errors. This project automates the decision-making process using machine learning techniques to improve efficiency and accuracy.

The system analyzes various applicant features such as income, loan amount, credit score, and other attributes to predict loan eligibility. It helps institutions minimize risk by identifying potential defaulters.

1.1 Dataset Description

The dataset used in this project is Loan_Default_noisy.csv, which contains applicant financial and demographic data.

Key Features:

- Loan Amount
- Income
- Credit Score
- Other categorical features (encoded using one-hot encoding)

Target Variable:

- **Status** → (Eligible / Not Eligible)

Preprocessing Steps:

- Missing values handled using:
  - Mean (for numerical data)
  - Mode (for categorical data)
- Categorical features converted using:
  - `pd.get_dummies()`
- Feature scaling using:
  - `StandardScaler`
- Class imbalance handled using:
  - RandomOverSampler

1.2 Flow Diagram

Fig.1 flow diagram.

2. Modules for Project Implementation

2.1 Modules Used

1. Pandas

Used for data handling

- `read_csv()` – Load dataset
- `fillna()` – Handle missing values

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2. NumPy

Used for numerical operations

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3. Matplotlib

Used for visualization

- ROC Curve
- Bar graphs
- Feature importance

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4. Scikit-learn

Core ML library

Functions used:

- `train_test_split()`
- `StandardScaler()`
- Models:
  - SVM (Linear + RBF)
  - KNN
  - Random Forest
- Metrics:
  - Accuracy
  - F1 Score
  - ROC-AUC
  - Recall
  - Confusion Matrix

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5. Imbalanced-learn (imblearn)

- `RandomOverSampler()` → Handles class imbalance

2.2 Platform Used

Jupyter Notebook

Used for:

- Running ML code
- Visualization
- Experimentation

Python

Chosen for:

- Simplicity
- Strong ML ecosystem

3. Machine Learning Model/ Searching Technique Used(Describe each ML model/ searching technique used)

- In this project, multiple machine learning algorithms were implemented to predict loan eligibility. Each model was trained on the preprocessed dataset and used to analyze patterns in applicant data.

- 1. Linear Support Vector Machine (Linear SVM)

  Linear SVM is a supervised machine learning algorithm used for classification. It works by finding a straight-line decision boundary (hyperplane) that separates data into different classes.

- Use in Project:
  In this project, Linear SVM was applied to classify loan applicants based on their features. It was used to understand how well a linear decision boundary can separate eligible and non-eligible applicants.

- 2. K-Nearest Neighbors (KNN)

  KNN is a non-parametric algorithm that classifies a data point based on the majority class of its nearest neighbors in the dataset.

- Use in Project:
  KNN was used to predict loan eligibility by comparing each applicant with similar applicants in the dataset. It helped analyze how local patterns in the data influence prediction.

- 3. Random Forest

  Random Forest is an ensemble learning algorithm that combines multiple decision trees to improve prediction accuracy and reduce overfitting.

- Use in Project:
  In this project, Random Forest was used to train the model on the dataset and capture complex relationships between features such as income, loan amount, and credit score. It helped in making more robust and reliable predictions.

- 4. Support Vector Machine (RBF Kernel)

  SVM with RBF kernel is used for non-linear classification problems. It transforms the data into a higher-dimensional space to find a non-linear decision boundary.

- Use in Project:
  This model was used to handle non-linear patterns in the dataset. It allowed better classification of loan applicants when the relationship between features was not linear.

- Conclusion of Model Usage

  All the above models were implemented and trained on the dataset to compare their ability to predict loan eligibility. Each model provided different insights into the data, helping in selecting the most suitable approach for the problem.

4. Evaluation metrics used (Describe each evaluation metric used along with formulae)

1. Accuracy

Accuracy = Correct Predictions / Total Predictions

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2. F1 Score

Balances precision and recall

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3. ROC-AUC Score

Measures model's ability to distinguish classes

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4. Recall

Important for detecting defaulters
(How many actual defaulters were correctly identified)

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5. Confusion Matrix

Shows:

- **True Positive**
- **False Positive**
- **True Negative**
- **False Negative**

5. Results and Visualizations

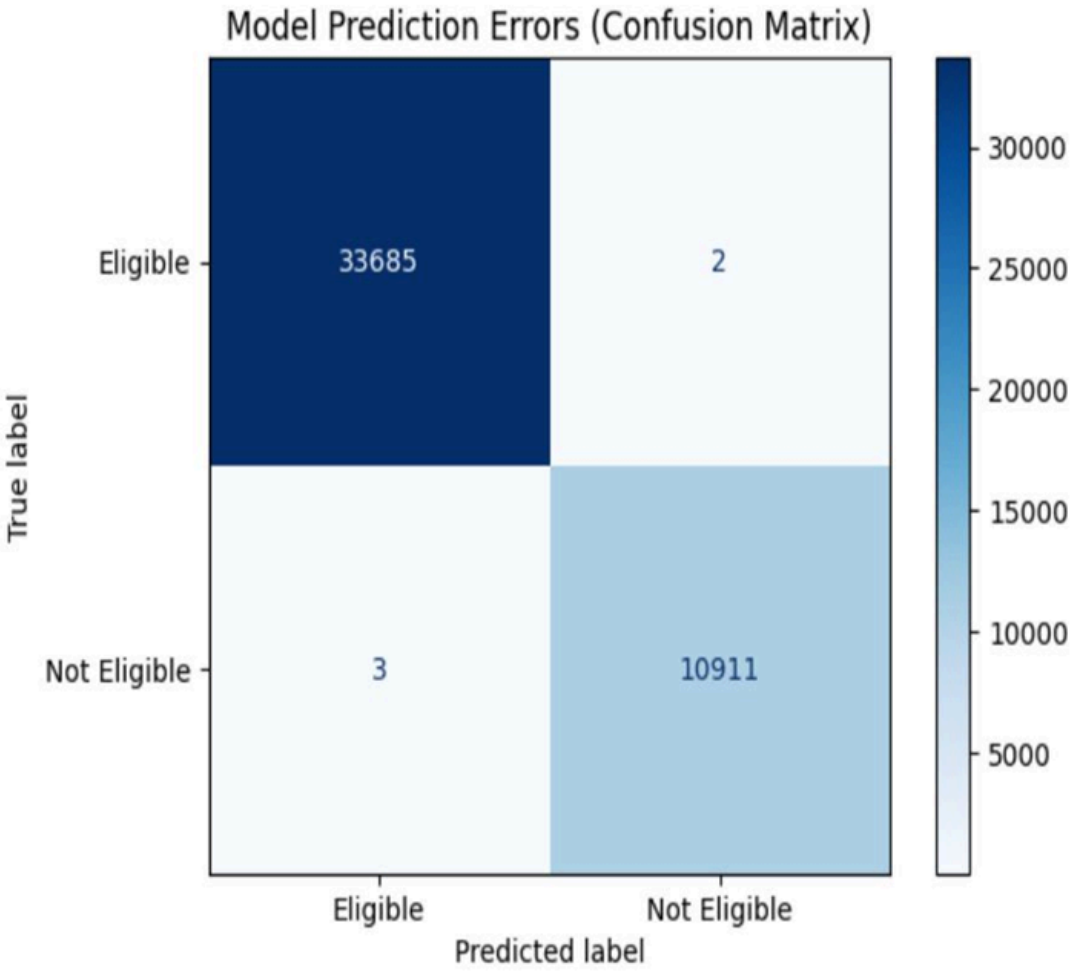

Fig.2 Confusion Matrix

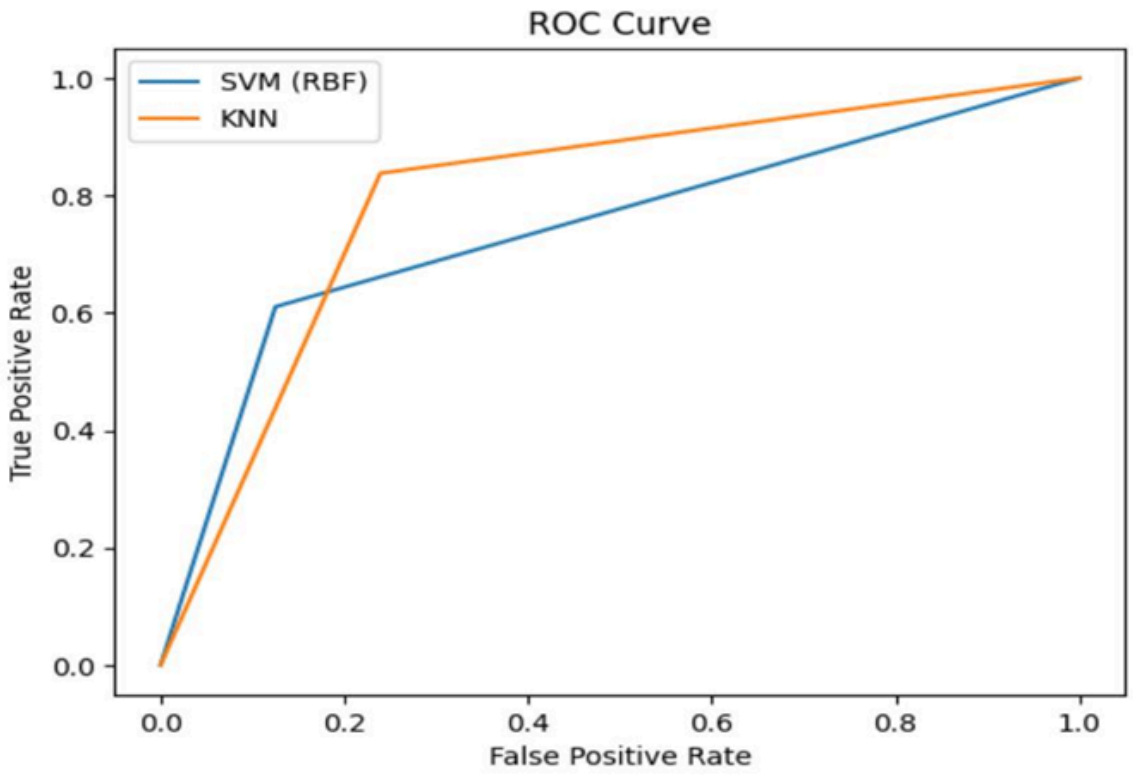

Fig.3 ROC Curve

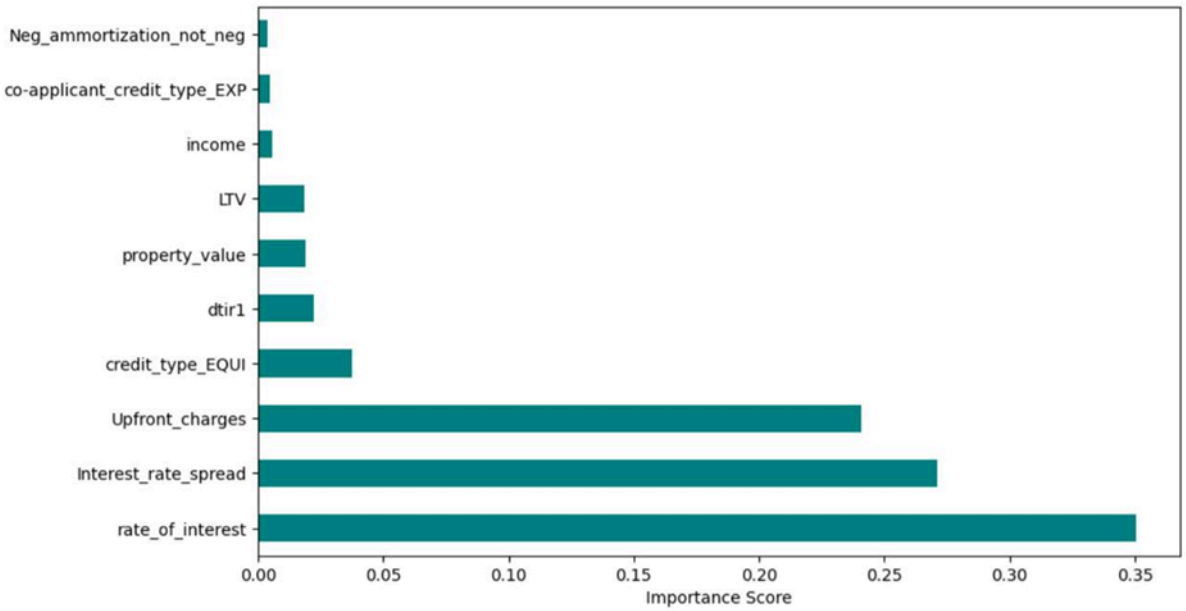

Fig.3 Importance score

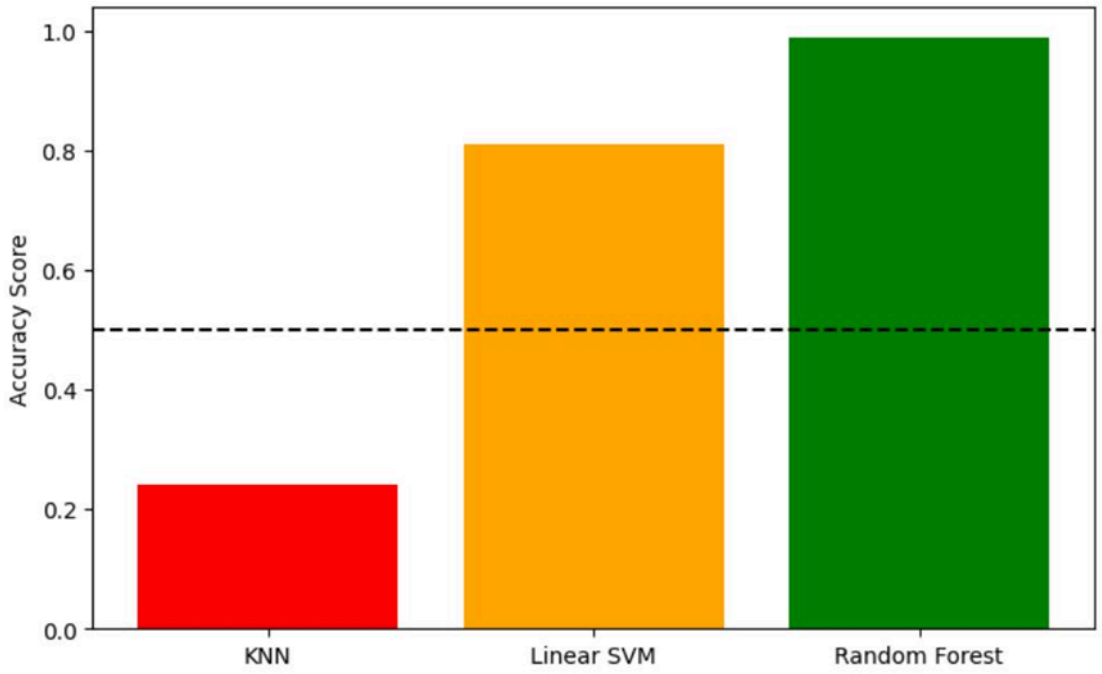

Fig.4 Accuracy score vs KNN,Linear SVM and Random Forest

	Model	Accuracy	F1-Score	ROC-AUC	Recall	Parameters
0	Linear SVM	0.810228	0.611529	0.742688	0.610409	C=1.0, Max Iter=1000
1	KNN	0.779557	0.650356	0.799251	0.837823	n_neighbors=5, weights='uniform'
2	Random Forest	0.999888	0.999771	0.999833	0.999725	n_estimators=100, max_depth=None
3	SVM RBF	0.938611	0.887390	0.955459	0.988455	C=1.0, kernel='rbf'

Fig5. Model accuracy in tabular form

6. Conclusion and Future Scope:

Conclusion

The project successfully predicts loan eligibility using multiple machine learning models. It improves decision-making accuracy and reduces financial risk for institutions.

Future Scope

- **Deploy as web application**
- **Use deep learning models**
- **Integrate with real-time banking systems**
- **Improve dataset quality for better predictions**

7. References:

Books:

- Let Us Python – Yashavant Kanetkar

Online:

- GeeksforGeeks
- Stack Overflow
- Scikit-learn documentation